

Compliance Matrix: ISO270001 and NIST CSF

This compliance matrix maps the implemented AWS cloud security controls of the Secure Cloud Banking Application project with ISO 27001 and NIST Cybersecurity Framework (NIST CSF) requirements. The matrix shows how different AWS security services and configurations help in maintaining cloud security and compliance.

1. ISO 27001 Compliance Matrix

A	B	C	D	E
		ISO 27001 Mapping		
Implemented Security Control	AWS Service / Feature	ISO 27001 Control	Compliance Purpose	
Identity and Access Management	AWS IAM	A.9 Access Control	Manage user authentication and least privilege access	
Data Encryption	AWS KMS, SSE-KMS	A.10 Cryptography	Protect sensitive banking data using encryption	
Network Security	VPC, Security Groups, NACL	A.13 Network Security	Secure communication and restrict unauthorized access	
Monitoring and Logging	CloudWatch, CloudTrail	A.12 Logging & Monitoring	Track activities and detect suspicious events	
Threat Detection	AWS GuardDuty, Security Hub	A.16 Incident Management	Identify and respond to threats quickly	
Backup and Recovery	Amazon S3, RDS Snapshots	A.17 Business Continuity	Maintain backup and disaster recovery	
Incident Response Plan	Cloud Incident Response Process	A.16 Information Security Incident Management	Handle security incidents effectively	
Secure Web Protection	AWS WAF, AWS Shield	A.13 Network Protection	Protect application from web attacks and DDoS	

2. NIST Cybersecurity Framework (NIST CSF) Compliance Matrix

C14				
	A	B	C	D
1			NIST CSF Mapping	
2	NIST Function	Implemented AWS Security Control	AWS Services Used	Purpose
3	Identify	Asset and identity management	IAM, VPC	Identify users, roles, and cloud resources
4	Protect	Encryption and access control	KMS, IAM, Security Groups	Protect data and systems from unauthorized access
5	Protect	Web Application Protection	AWS WAF, AWS Shield	Prevent web attacks and DDoS attacks
6	Detect	Threat monitoring and alerting	CloudWatch, GuardDuty, Security Hub	Detect suspicious cloud activities
7	Respond	Incident response handling	CloudTrail Logs, SOC Process	Respond quickly to cloud security incidents
8	Recover	Backup and disaster recovery	Amazon S3, RDS Snapshots	Restore services and data after incidents
9				

Conclusion

The Secure Cloud Banking Application project follows important security practices aligned with ISO 27001 and the NIST Cybersecurity Framework. AWS services like IAM, KMS, VPC, CloudTrail, GuardDuty, WAF, and S3 help in implementing cloud security, monitoring, threat detection, and incident response. This compliance matrix demonstrates how the project supports secure cloud operations and industry-standard security requirements.